Lorentz-Covariant Scalar Field Model:

Effective Field Theory Realization of Post-Newtonian Gravity

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Abstract

We present a Lorentz-covariant complex scalar field model whose weak-field limit reproduces the post-Newtonian predictions of General Relativity (GR). The scalar coherence field $C = |\psi|^2$ provides an effective field-theoretic realization whose coarse-grained limit yields the standard Poisson sector with $\gamma_{PPN} = 1$ in static regimes. The model introduces no preferred spatial frame; ψ is a Lorentz-covariant scalar field minimally coupled to $g_{\mu\nu}$, with all dynamics deriving from a standard covariant action. We refer to this framework as TRR (Theory of Rendered Reality) for shorthand.

A complex scalar field $\psi(x,t)$ oscillates at carrier frequency ω_c (constrained: $\omega_c > 1.4 \times 10^{15}$ rad/s from optical clock synchronization) and couples to matter through phenomenological coupling g_m [units: m^3 kg^{-1} s^{-2}]. Under the slowly-varying envelope approximation (SVEA) with static sources and $L \gg \lambda_c$, the framework produces $\nabla^2(\delta C) = -\kappa\rho$ with $\kappa = 4\pi G/c^2$ [units: m kg^{-1}], matching tested post-Newtonian parameters $(\gamma = 1, \beta \approx 1)$ within experimental precision.

Current observational constraints:

Observable	Constraint
Cassini γ_{PPN}	$1 + (2.1 \pm 2.3) \times 10^{-5}$
Lunar Laser Ranging β_{PPN}	≈ 1
GW150914 ringdown	QNM $\ell = 2, m = 2$ precision
GW170817 tidal Love	Bounds on k_2
Optical clock (Setup 3)	$\epsilon < 1.49 \times 10^{-6}$ (3σ)

The allowed parameter space: $10^{-10} < \epsilon < 1.49 \times 10^{-6}$, $\kappa_{max} > 0.95$, $\omega_c > 1.4 \times 10^{15}$ rad/s. The parameters (g_m, λ, ϵ) are treated phenomenologically; their ultraviolet origin is not specified.

I. Introduction

1.1 Scalar Effective Field Theory Framework

General Relativity describes the geometry of spacetime—curvature, geodesics, gravitational wave propagation—as a complete classical theory of gravity. We present an alternative effective field theory formulation based on a Lorentz-covariant complex scalar field $\psi(x,t)$ that reduces to GR's predictions in tested regimes while providing distinct strong-field signatures.

The scalar field carries a coherence amplitude $C = |\psi|^2$ that couples to matter distributions. Mass creates coherence perturbations $\delta C \equiv C - \Psi_0^2$, and these perturbations satisfy:

$$\nabla^2(\delta C) = -\kappa\rho, \quad \kappa = 4\pi G/c^2$$

Critical clarification on Lorentz covariance: The model introduces no preferred spatial frame. The scalar field ψ is a Lorentz-covariant field minimally coupled to the metric $g_{\mu\nu}$. Observable Lorentz-violating effects are suppressed by $(\lambda_c/L)^2 \ll 1$ in all tested regimes. Unlike pre-relativistic ether models, the substrate does not define a measurable preferred rest frame.

1.2 Covariant Action

All dynamics derive from a standard covariant action:

$$S = \int d^4x\sqrt{-g}\left[-\frac{1}{2}(\partial_\mu\psi^*)(\partial^\mu\psi) - \frac{\omega_c^2}{2c^2}|\psi|^2 - V(|\psi|^2) - g_m\rho|\psi|^2\right]$$

Mexican hat potential: $V(|\psi|^2) = (\lambda/4)(|\psi|^2 - \Psi_0^2)^2$. Variation $\delta S/\delta\psi^* = 0$ yields:

$$\Box\psi + \frac{\omega_c^2}{c^2}\psi + \frac{\partial V}{\partial|\psi|^2}\psi = -g_m\rho\psi$$

1.3 Test Particle Motion from Emergent Metric

Macroscopic test particles follow geodesics of the emergent effective metric, not a direct scalar force. The matter coupling $-g_m\rho|\psi|^2$ sources coherence perturbations δC via the Poisson equation, which in turn define the metric perturbation.

Weak-field metric identification: The framework identifies:

$$h_{00} = -2\delta C$$

where h_{00} is the time-time component of the metric perturbation in $g_{00} = -1 + h_{00}$.

Geodesic acceleration: In the weak-field, slow-motion limit, test particle acceleration follows from the geodesic equation:

$$\mathbf{a} = \frac{c^2}{2} \nabla h_{00}$$

Substituting the coherence-metric identification: Using $h_{00} = -2\delta C$:

$$\mathbf{a} = \frac{c^2}{2} \nabla(-2\delta C) = -c^2 \nabla \delta C$$

Point-mass solution: From the Poisson equation $\nabla^2(\delta C) = -\kappa\rho$ with $\kappa = 4\pi G/c^2$, the solution for a point mass M is:

$$\delta C(r) = \frac{GM}{c^2 r}$$

Computing the gradient (pointing radially inward):

$$\nabla \delta C = -\frac{GM}{c^2 r^2} \hat{r}$$

Final result: Substituting into the acceleration equation:

$$\mathbf{a} = -c^2 \times \left[-\frac{GM}{c^2 r^2} \hat{r} \right] = -\frac{GM}{r^2} \hat{r}$$

This exactly recovers Newtonian gravity. Test particles follow geodesics of the emergent metric, accelerating toward regions of higher coherence (δC increases toward mass).

II. Parameter Definitions

2.1 Phenomenological (Primary) Parameters

Terminology note: In the present work, "coherence" refers strictly to the scalar field amplitude squared $C = |\psi|^2$ and carries no additional quantum-mechanical interpretation.

These parameters are inputs constrained by observation. The parameters (g_m, λ, ϵ) are treated phenomenologically; their ultraviolet origin is not specified.

Symbol	Definition	Units	Constraint
g_m	Matter-substrate coupling	$\mathrm{m}^3\ \mathrm{kg}^{-1}\ \mathrm{s}^{-2}$	$g_m \Psi_0^2 = 2\pi G = 4.19 \times 10^{-10}\ \mathrm{m}^3\ \mathrm{kg}^{-1}\ \mathrm{s}^{-2}$ (CODATA)
λ	Substrate self-coupling (Mexican hat)	Dimensionless	$\lambda \sim 10^{-2}$. Phenomenological parameter.
ϵ	Mode-transition proximity: $\epsilon \equiv \Delta r / r_s$	Dimensionless	$\epsilon < 1.49 \times 10^{-6}$ (Setup 3, 3σ). Phenomenological parameter.

2.2 Derived Parameters

Symbol	Definition	Units	Value
κ	Gravitational coupling: $\kappa = 4\pi G/c^2$	m kg^{-1}	9.33×10^{-27} m kg^{-1} (CODATA). No free parameters.
r_c	Mode-sector transition: $r_c = r_s(1+\epsilon)$	m	Depends on M and ϵ
κ_{max}	Stability margin parameter. $\kappa_{max} \equiv 1/(1+\delta C_{max})$. $\kappa_{max} \to 1$ corresponds to $\delta C_{max} \to 0$ (GR-like).	Dimensionless	$\kappa_{max} > 0.95$. This is a stability parameter, NOT a maximum coherence value.
ω_c	Substrate carrier frequency	rad/s	$\omega_c > 1.4 \times 10^{15}$ rad/s (optical clock synchronization)

III. Post-Newtonian Consistency

3.1 PPN Parameter Recovery

The present scalar field model matches tested post-Newtonian parameters $(\gamma = 1, \beta \approx 1)$ within experimental precision under the explicit assumptions of SVEA, static sources, and $L \gg \lambda_c$.

PPN γ parameter:

$$\gamma_{PPN} = 1.000\ldots \ \text{(exact)}$$

Cassini tracking yields $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti et al. 2003).

PPN β parameter:

$$\beta_{PPN} = 1 + O(\lambda(\delta C)^2) \lesssim 10^{-20}$$

Lunar Laser Ranging constrains $\beta - 1 = (1.2 \pm 1.1) \times 10^{-4}$ (Nordtvedt 1968; Williams et al. 2004).

3.2 Solar System Tests

Test	GR Prediction	Present Model	Agreement
Light deflection	1.7506″ (Sun)	Identical ($\gamma = 1$)	Exact
Perihelion precession (Mercury)	43.0″/cy	Identical	Exact
Shapiro delay	Viking result	Identical ($\gamma = 1$)	Exact

IV. Strong-Field Mode-Sector Structure

4.1 Mode-Transition Proximity Parameter ϵ

The parameter $\epsilon \equiv \Delta r/r_s = (r_c - r_s)/r_s$ is defined geometrically as the fractional radial gap between the mode-sector transition radius r_c and the Schwarzschild radius r_s.

Physical interpretation:

- $\epsilon = 0$: Mode transition at r_s → GR limit
- $\epsilon \sim 10^{-10}$ to 10^{-6}: Small gap, deviations highly suppressed
- $\epsilon \sim 0.1$: Large gap → ruled out

4.2 Observational Constraints on ϵ

Optical clock synchronization (Setup 3):

$$\epsilon < 1.49 \times 10^{-6} \text{ (3}\sigma\text{ confidence)}$$

This represents a 6.7× improvement over orbital bounds. **This bound assumes linear environmental regression and independence of clock residuals; correlated systematic effects may weaken the constraint.**

GW150914 ringdown: $\epsilon < 10^{-10}$ (conservative WKB estimate, model-dependent)

GW170817 tidal deformability: $\epsilon < 10^{-6}$ to 10^{-7}

4.3 WKB Approximation and Smoothing Corrections

The quasi-normal mode shift estimate uses a sharp-transition WKB approximation:

$$\delta\omega/\omega \sim \pi/\ln(1/\epsilon)$$

In practice, the transition has finite width $\Delta r_{smooth} \sim r_s/\sqrt{\ell(\ell+1)}$, introducing ~40% corrections for $\epsilon \sim 10^{-10}$ and $\ell = 2$. **The QNM bound is therefore shape-dependent and requires numerical validation.**

Model-dependence summary:

Observable	Robustness	Notes
Echo delay	Robust	Determined by tortoise-coordinate log. Smoothing gives additive corrections.
QNM shift	Not robust	Depends on barrier shape. Requires numerical solution.
Echo amplitude	Moderately sensitive	Smoothing changes integrand but exponential suppression likely preserved.
Love numbers	Depends on matching	Could alter prefactor but likely preserves ϵ^2 scaling if $\epsilon \ll 1$.

V. Strong-Field Observables and Predictions

5.1 Tidal Love Numbers

$$k_2 \propto \epsilon^2 \text{ for } \epsilon \ll 1$$

For $\epsilon \sim 10^{-6}$: $k_2 \sim 10^{-12}$, thirteen orders of magnitude below GR's $k_2^{GR} \sim 0.1$ for neutron stars.

5.2 Quasi-Normal Modes

$$\delta\omega/\omega \sim \pi/\ln(1/\epsilon)$$

The QNM shift estimates should be interpreted as order-of-magnitude scaling relations rather than precise frequency predictions.

For $\epsilon = 10^{-10}$: $\delta\omega/\omega \sim 0.14$ (14% frequency shift)

5.3 Gravitational Wave Echoes

$$\Delta t_{echo} = \frac{2r_s}{c}\ln(1/\epsilon)$$

For $M = 30M_\odot$ and $\epsilon = 10^{-6}$: $\Delta t_{echo} \sim 0.4$ ms

Echo amplitude: $R_{GW} \sim \exp\left(-2\int_{r_s}^{r_c} Im(k_\ell)dr^*\right) \sim 10^{-5}$ to 10^{-3}

5.4 Scalar Breathing Mode

$$f_s = \frac{\alpha}{\sqrt{\kappa_{max}}} f_{22}$$

where $\alpha \approx 3.8 \pm 0.2$ from numerical WKB fits. For typical parameters: $f_s \sim 4f_{22} \sim 1$ kHz

VI. Falsification Criteria

The framework is falsified by any of the following observations:

Observation	Implication
1. Multi-mode ringdown precision: Modes match Kerr at <0.1% uniformly	$\epsilon < 10^{-12}$, extreme fine-tuning required
2. Stacked echo detection: $R > 0.01$ at $\Delta t = 10$–100 ms	Contradicts exponential suppression for $\epsilon < 10^{-6}$
3. Love number detection: $k_2 > 10$ measured in NS-NS merger	Contradicts ϵ^2 suppression
4. Breathing mode observation: Scalar monopole at $f_s \sim 4f_{22}$ with SNR > 5	Confirms prediction
5. PPN violations: $\lvert\gamma - 1\rvert > 10^{-4}$ measured	Weak-field consistency broken. Model excluded.

VII. Observational Status and Future Tests

7.1 Domain of Validity

Static and weak-field ($L \gg \lambda_c, |\Phi/c^2| \ll 1$): Framework recovers GR post-Newtonian predictions exactly.

Strong-field exterior ($r > r_s$): Schwarzschild and Kerr geometries preserved in $\epsilon \to 0$ limit. Observable deviations appear only through mode-sector boundary at $r_c = r_s(1+\epsilon)$, not through horizon modifications.

Interior modifications ($r < r_s$): Vortex-core dynamics confined behind horizon. All testable signatures derive from $r_c > r_s$.

7.2 Current Constraint Summary

Observable	GR	Present Model	Current Constraint
γ_{PPN}	1 (exact)	1.000... (exact)	$\lvert\gamma - 1\rvert < 2.3 \times 10^{-5}$ (Cassini)
β_{PPN}	1 (exact)	$1 + 10^{-20}$	$\lvert\beta - 1\rvert < 1.1 \times 10^{-4}$ (LLR)
GW speed	c (exact)	c (exact)	$\lvert v/c - 1\rvert < 10^{-15}$ (GW170817)
Love k_2	0.05–0.15	$\sim 10^{-12}$	$k_2 < 800$ (GW170817)
QNM f_{22}	251 Hz	$251 \pm \delta f$ Hz	$\delta f/f < 3\%$ (GW150914)
Echo amplitude	None	$R_{GW} \sim 10^{-5}$	$R < 0.01$ (stacked)
Breathing f_s	None	$\sim 4f_{22}$	Not detected (p=0.20)
Optical clock ϵ	N/A	Phenomenological	$\epsilon < 1.49 \times 10^{-6}$ (3σ)

7.3 Limiting Regimes

The framework contains four controlled limits that recover known physical regimes:

Limit	Physical Regime
1. $\lambda \to 0$	Framework approaches linear scalar-tensor theory
2. $\epsilon \to 0$	Mode-transition at horizon. All strong-field deviations vanish. Recovers GR exactly.
3. $\delta \to 0$	GR-stable basin boundary
4. Large r	$\delta C \to 0$, recovering Minkowski spacetime. Schwarzschild $1/r$ falloff preserved.

VIII. Discussion

8.1 Relationship to General Relativity

The present model provides an effective scalar field realization compatible with GR in tested regimes. In the weak-field limit, both theories yield identical predictions. As $\epsilon \to 0$, all strong-field deviations vanish and the theories converge asymptotically.

Ontological distinction: GR describes gravity via spacetime curvature (Einstein tensor $G_{\mu\nu}$ sourced by stress-energy $T_{\mu\nu}$). The present model describes gravity via coherence-defined effective potentials (δC perturbations sourced by matter density ρ). These are complementary descriptions at different scales—GR provides macroscopic geometric structure, while the scalar field model describes substrate dynamics.

8.2 Scalar Field Interpretation

This is not a revival of pre-relativistic ether models.

Three critical distinctions:

- **1. Lorentz covariance:** The scalar field is Lorentz covariant. Observable Lorentz-violating effects are suppressed by $(\lambda_c/L)^2 \sim 10^{-40}$ for astrophysical baselines.
- **2. Non-zero vacuum:** The Mexican hat potential stabilizes coherence at Ψ_0 rather than zero.
- **3. Physical mechanism:** Mass creates coherence enhancements ($\delta C > 0$ near mass), and test particles accelerate toward regions of higher coherence via the effective potential $U \propto \delta C$.

8.3 U(1) Symmetry and Helical Nonlinearity

The U(1) phase symmetry $\psi \to \psi e^{i\theta}$ uniquely selects the nonlinear interaction $g|\psi|^2\psi$. In the strong-field regime this produces quantized vortices, coherence solitons, and vortex core interiors.

Critical: The vortex-core interior dynamics do not modify exterior multipole moments or null geodesic structure outside $r = r_s$. All observational signatures derive from the mode-sector boundary at $r_c > r_s$, not from horizon modifications.

IX. Conclusions

1. A Lorentz-covariant complex scalar field model is presented whose weak-field limit matches tested post-Newtonian parameters ($\gamma = 1, \beta \approx 1$) within experimental precision under explicit assumptions (SVEA, static sources, $L \gg \lambda_c$).
2. Test particle motion is derived via the emergent spacetime metric. The matter coupling term sources coherence perturbations δC via the Poisson equation, which define the metric perturbation $h_{00} = -2\delta C$. Macroscopic test particles follow geodesics of this emergent metric, recovering Newtonian acceleration $\mathbf{a} = -GM/r^2\hat{r}$ in the weak-field limit through the relation $\mathbf{a} = -c^2\nabla\delta C$.
3. The mode-transition parameter $\epsilon \equiv \Delta r/r_s$ is defined geometrically. All strong-field observables are functions of ϵ alone at fixed mass. The parameters (g_m, λ, ϵ) are treated phenomenologically.
4. Optical clock analysis (Setup 3) yields $\epsilon < 1.49 \times 10^{-6}$ (3σ). **This bound assumes linear environmental regression and independence of clock residuals.**
5. Strong-field predictions: $k_2 \propto \epsilon^2$; $\delta\omega/\omega \sim \pi/\ln(1/\epsilon)$ (WKB estimate with ~40% smoothing corrections requiring numerical validation); echo amplitude $R_{GW} \sim 10^{-5}$; breathing mode $f_s = (3.8/\sqrt{\kappa_{max}})f_{22}$. All consistent with current null detections.
6. Four limiting cases ($\epsilon \to 0, \lambda \to 0, \kappa_{max} \to 1, \omega_c \to \infty$) each recover known physical regimes, confirming the theory is constrained rather than tunable.
7. Current constraints from Cassini γ_{PPN}, Lunar Laser Ranging β_{PPN}, GW150914 ringdown, GW170817 tidal Love numbers, and optical clock data define allowed parameter space: $10^{-10} < \epsilon < 1.49 \times 10^{-6}$; $\kappa_{max} > 0.95$; $\omega_c > 1.4 \times 10^{15}$ rad/s.
8. Next-generation detectors (Einstein Telescope, Cosmic Explorer) substantially increase sensitivity to predicted signatures and could either further constrain or exclude portions of parameter space within 10–15 years.
9. The framework provides an effective scalar field realization compatible with GR in tested regimes, offering complementary descriptions at different scales.

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- Garner, T.L. (2026b). Nonlinear Compact Object Solutions in Coherence Gradient Gravity (Paper II). *In preparation.*
- Garner, T.L. (2026c). Vortex Core Interiors and Gradient-Activated U(1) Breaking (Paper III). *In preparation.*

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